

Fictional case study for training. Not a real investigation.



Investigation report

Rejected take-off and runway overrun, Tarlton Springs - NTSB-style
probable cause report

Drafted 05/10/2026

INVESTIGATION WORKFLOW SUITE

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Executive summary

What happened

On 18/06/2026 at 17:30 local time, Saltbush Air SB512 began its take-off from runway 09 at Tarlton Springs Aerodrome in darkness. The aircraft was VH-SBQ, a 50-seat twin-turboprop. As it passed V1, the first officer saw two kangaroos crossing the runway from the northern side. The pilot in command rejected the take-off at 118 kt, 6 kt above V1. One second later the No. 2 propeller and right main landing gear struck both animals. The aircraft left the end of the runway at approximately 21 kt. It stopped 62 m into the 90 m runway end safety area, upright on the extended centreline.

There were 44 passengers, 2 flight crew and 1 cabin crew on board. No evacuation was commanded, and passengers left by the forward airstair. There were no serious injuries. One cabin crew member reported bruising during the disembarkation. The aircraft was substantially damaged. Three No. 2 propeller blades were beyond limits, the right main gear doors were distorted, both right main tyres deflated through the fuse plugs, and the lower right fuselage was punctured. Both kangaroos were killed. The aerodrome closed for 4 hours 20 minutes.

What the investigation found

The probable cause was a take-off rejected 6 kt above V1 when two kangaroos moved onto the runway, combined with the loss of No. 2 reverse thrust in the strike that followed. The scheduled stopping margin was 67 m. Starting the stop late consumed that margin. The strike then reduced deceleration to approximately 15% below the certified figures. Both were needed to carry the aircraft off the end, and neither alone would have done so.

The northern perimeter fence had been recorded as failed since 04/2023, with animals seen passing through it. Its repair was deferred as unfunded at three budget rounds. The breach was very likely the animals' route onto the runway. The departure was in the hour after last light, when the aerodrome's own log recorded most kangaroo movement. No runway inspection was requested before the roll. The operator required one only at aerodromes listed after a reported strike. Tarlton Springs had eleven logged sightings of kangaroos on the sealed surface but no strike, so it was never listed. The evening service had been approved as a schedule change, and the route risk assessment restricting the aerodrome to daylight was not reopened. The aerodrome also fell below the size threshold for a documented wildlife plan. That likely contributed, although it is held with less confidence than the other findings.

Recurrent training presented only aircraft system failures, and this likely contributed to the reject decision. The pilot in command stated that he departed from the rule deliberately. The training gap therefore explains the decision without excusing it. Maximum braking was established 2 seconds after the decision, faster than the fleet mean of 3.2 seconds, and this very likely limited the overrun.

Two dark floor proximity lighting segments and an 11-hour, five-sector duty were each unlikely to have influenced the occurrence, but both increased risk. Edge lights in the section the animals crossed measured above nominal seven days later. Nothing recorded their output on the night, and the two pilots' accounts differed. Whether the lighting affected detection of the animals could not be determined. A tailwind and a No. 2 engine power loss before the reject were ruled out.

What can be done

A control that is armed only by a reported strike cannot prevent the first one. On 26/06/2026 the operator listed Tarlton Springs by temporary revision TR-2026-04, so an inspection is now required before every departure there after last light. That change covers one aerodrome only. The

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investigation recommends that the operator list aerodromes on wildlife exposure rather than on reported strikes. It also recommends a route risk assessment trigger for any aerodrome first operated into after last light. The Head of Flying Operations suspended the evening service pending the investigation.

The aerodrome operator installed a temporary electric wire on 25/06/2026. It has programmed replacement of the 140 m fence section for 09/2026. No action had been recorded on the deferral of a known defect, the wildlife plan threshold, the change approval process, obstruction training, or the monitoring of floor proximity lighting across the type.

Factual information

History of the flight

All times are UTC. The duty record places the occurrence at 17:30 local time (UTC plus 10 hours).

On 18/06/2026 Saltbush Air SB512 was the last scheduled sector of the day from Tarlton Springs Aerodrome. The only runway inspection that day was at 21:35 on 17/06/2026, before the first scheduled arrival, and recorded the movement area as clear. No further inspection was carried out or requested.

The doors closed at 07:16, 40 minutes behind schedule, with 44 passengers, 2 flight crew and 1 cabin crew on board. The aerodrome was in darkness. The first officer was pilot flying.

At 07:29:10 the tower cleared SB512 for take-off on runway 09, wind 090 at 6 kt. Neither the tower nor the crew raised a runway inspection. No wildlife transmission appears on the tower channel in the preceding 90 minutes.

Power was set at 07:30:02. The pilot in command called "Power set, both engines good" at 07:30:04 and V1 at 07:30:29 at 112 kt. At 07:30:31 the first officer called "Roos - roos, roos, on the left, they're on the -". The occurrence notification records that the animals moved onto the runway from the northern side, past the 1,100 m point. The first officer estimated the range at first sight at about 200 m.

At 07:30:32 the pilot in command called "Stopping. I have control." The take-off was rejected at 118 kt, 6 kt and 3 seconds after V1, 1,152 m along runway 09 with 695 m of sealed surface remaining. Both power levers reached flight idle within 0.6 seconds.

At 07:30:33 the right main landing gear and the No. 2 propeller struck both animals. Maximum braking was established at 07:30:34. At 07:30:36 the pilot in command called "Reverse - number two's not coming. Max brakes, max brakes." Reverse was achieved on No. 1 only. At 07:30:47 the pilot in command said "We're not stopping on it."

The aircraft left the end of runway 09 at 07:30:51 at approximately 21 kt. It came to rest at 07:30:57, 62 m into the 90 m runway end safety area (RESA), upright on the extended centreline with the nose wheel in graded gravel.

The pilot in command set the parking brake, and both engines were shut down at 07:31:26. There were no fire indications. The pilot in command stated "I'm not evacuating", and the first officer agreed. The cabin was told to remain seated. At 07:31:54 the pilot in command requested activation of the emergency plan and a vehicle. The tower activated the aerodrome emergency plan at 07:32:10, and aerodrome staff attended with a vehicle. There was no fire.

The forward airstair was heard operating at 07:34:40, and passengers disembarked by it. No evacuation was commanded and no slides were deployed. The occurrence notification records disembarkation 4 minutes 8 seconds after the aircraft stopped. The first officer recalled that the main cabin lights stayed on and the disembarkation was orderly.

No serious injuries occurred. One cabin crew member reported bruising during the disembarkation. Both flight crew returned negative drug and alcohol tests.

Personnel information

Pilot in command

The pilot in command held 8,410 hours total, 2,140 on type, and had been in command for 3 years 1 month. The last cyclic simulator check (11/03/2026) and line check (02/12/2025) were satisfactory.

There was no prior occurrence history. Duty and rest in the preceding 7 days were within company limits, with no variations. The pilot in command described having two rest days before the duty.

First officer

The first officer held 1,655 hours total, 405 on type, and had been with the company for 14 months. The last cyclic simulator check, on 28/04/2026, was satisfactory on all items. The first officer had flown two prior sectors into Tarlton Springs, on 09/01/2026 and 17/02/2026, both in daylight. This was the first officer's first operation there in darkness. The first officer described having one rest day before the duty.

Duty on 18/06/2026

The crew signed on at 06:30 local for five sectors. The occurrence sector, the fifth, had been added that morning when the rostered crew reported unfit. The occurrence was 11 hours 00 minutes into a planned duty of 12 hours 05 minutes, ending at 18:35 local. The maximum permitted duty for five sectors with this sign-on was 12 hours 30 minutes. Both pilots described themselves as tired but not struggling.

Training

Both pilots stated that every stop or go scenario in their recurrent simulator training had been an engine problem or a configuration warning below V1. Neither recalled any scenario involving a runway obstruction. The first officer stated that the standard take-off brief assigns the reject decision to the captain below V1 and calls for continuing above V1. It does not address a runway obstruction.

Aircraft and equipment information

VH-SBQ was a 50-seat twin-turboprop. Take-off was planned at 20,800 kg, flap 10, dry runway 09, with a 5 kt headwind component.

The post-occurrence inspection found:

- No. 2 propeller: three blades damaged beyond repairable limits, with leading edge deformation and one blade tip separation. Animal tissue was recovered from two blade roots. The engine required a shock-load inspection.
- Engines: borescope inspection found no pre-existing damage and no evidence of a power anomaly before the impact.
- Right main landing gear: both tyres deflated through the fuse plugs, consistent with brake energy. The gear doors were distorted by the impact. The gear structure was within limits.
- Brakes: absorbed energy was calculated at 84% of the certified maximum for the weight and speed. There was no brake fire and no hydraulic leak. The anti-skid system tested serviceable.
- Lower right fuselage: abrasion and two punctures consistent with propeller debris.

The first officer reported seeing unlit sections of the aisle floor proximity lighting strip towards the rear of the cabin during the disembarkation, with emergency lighting armed and ceiling lights on, and raised this with an engineer the next morning. A functional check was carried out on 24/06/2026. [NOT ESTABLISHED: result of the 24/06/2026 functional check of the floor proximity lighting]

Meteorological information

The recorded wind gave a headwind component of 4 to 6 kt throughout the roll, consistent with the 07:00 routine observation. No tailwind component was recorded. Runway 09 was dry. There was no town lighting near the aerodrome. [NOT ESTABLISHED: time of last light at Tarlton Springs on 18/06/2026]

Recorded data

The flight data recorder (FDR) and quick access recorder readout showed:

- Both engines at take-off torque, 98.4% and 98.5%, from power set until the power levers were retarded at 07:30:32.
- No exceedance, caution or warning before that point. Fuel flow, ITT and propeller RPM were within normal scatter.
- At impact, a lateral acceleration spike of 0.31 g right and a vertical spike of 1.48 g.
- No. 2 propeller RPM decaying from 07:30:33, and the No. 2 reverse interlock not making.
- Brake pressure of 2,940 psi at 07:30:34, with anti-skid active on all four wheels.

The decision-to-maximum-braking time was 2 seconds, at the fast end of the operator's fleet distribution (mean 3.2 seconds).

Mean deceleration from 07:30:34 to 07:30:44 was 2.6 m/s², approximately 15% below the certified accelerate-stop data for this weight and runway condition, which assume symmetric reverse. In the gravel, deceleration fell to 1.9 m/s². The stopping distance from 118 kt was 757 m.

The cockpit voice recorder (CVR) and tower audio captured the exchanges quoted above. After the aircraft stopped, the first officer said "They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less." The pilot in command said "I never had them. I was heads-in on the engine page after V one."

The operator's engineering representative removed the FDR and CVR at 12:40 on 18/06/2026.

Aerodrome and site information

The tower at Tarlton Springs was open and closed at 18:00 local as rostered. Runway 09 is followed by a 90 m RESA of graded gravel, beyond which the terrain is ungraded. The aircraft stopped 28 m short of the end of the prepared area.

The runway survey found:

- Two kangaroo carcasses, at 1,188 m and 1,221 m from the runway 09 threshold.
- Continuous tyre marks from 1,160 m to the runway end.
- A nose wheel rut extending 62 m into the RESA on the extended centreline.

The aerodrome's paper wildlife sighting log held 46 entries from 01/2025 to 06/2026:

- 41 recorded kangaroos on or adjacent to the movement area.
- 38 of those 41 were timed between one hour before and two hours after last light.
- 11 recorded animals on the sealed surface of runway 09/27.

The log was not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked for it.

The aerodrome did not meet the size threshold above which a documented wildlife plan was required. The perimeter fence repair had been deferred for three budget cycles. [NOT ESTABLISHED: condition of the perimeter fence and the size threshold applied]

The pilots differed on the runway lighting. The pilot in command described it as brighter than wanted, with black either side, but could not say whether this affected detection. The first officer

described it as normal but had no night experience there to compare it with, and stated that the animals were clearly visible once on the sealed surface. [NOT ESTABLISHED: runway edge lighting intensity setting in use]

Organisational and management information

The operator's Part A section 8.14 and Appendix 8A list aerodromes at which a runway inspection is requested before the first departure after dark. Listing is triggered by a strike, and Tarlton Springs was not listed. The flight plan, charts, ERSAs entry and company aerodrome brief page for Tarlton Springs, all reviewed by the first officer, made no mention of wildlife.

The evening Tarlton Springs service was approved as a schedule change, and the route risk assessment was not reopened. The pilot in command recalled that the service began around the middle of 2025, replacing a daylight service. Neither pilot recalled any accompanying briefing.

The pilot in command had seen kangaroos on the strip at Tarlton Springs in the evening four or five times and had mentioned this to ground staff, but had not filed a report. The first officer stated that ground staff described the animals as present every night.

Additional information

The aerodrome closed for 4 hours 20 minutes. Two kangaroos were killed. The Head of Flying Operations suspended the evening Tarlton Springs service pending the outcome of the investigation.

Analysis

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Section 10.4 of AS-139 sets two levels of obligation. Every certified aerodrome operator "must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome" (10.4.1). A documented wildlife hazard management plan is required only where the aerodrome is used by aeroplanes with a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year (10.4.2). Below both thresholds, the measures "need not be documented", although the operator "must be able to demonstrate what measures are in place" (10.4.3).

Tarlton Springs fell below both thresholds. Its heaviest scheduled aeroplane weighed 22,900 kg, and it handled 61,400 passenger movements in 2025. It accordingly held no plan, no hazard assessment and no dispersal programme. The only documented measure was the single inspection before the first arrival under Aerodrome Manual 6.3. These facts were confirmed against the certified aerodrome register and the aerodrome operator's compliance file. The aerodrome was not in breach of the standard. It was virtually certain that the threshold existed and that it placed Tarlton Springs outside the requirement for a documented plan.

The open question was whether that absence contributed. Two sets of information held at the aerodrome were relevant.

The first was the aerodrome's own sighting log for 01/2025 to 06/2026. It contained 46 entries. Forty-one recorded kangaroos on or adjacent to the movement area. Thirty-eight of those were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked for it.

The second was the council's works register. From 04/2023 it recorded the northern perimeter fence as failed, with animals observed passing through. The repair was deferred as unfunded at the 2023, 2024 and 2025 budget rounds.

The argument for influence was that a documented plan is the mechanism that would have linked the sighting log to the operators using the aerodrome. It would also have placed the failed fence on a funded programme rather than leaving it on an unfunded works register. Without a plan, each piece of knowledge stayed where it was first recorded.

The argument against was substantial. Section 10.4.1 still obliged the aerodrome operator to take reasonable measures, whether or not those measures were documented. A documented plan is not the only way to repair a fence or to pass a sighting log to an airline. The deferrals were budget decisions made by the council. There was no evidence that a documented plan would have attracted funding any more successfully. A plan sets out what should be done, but repairing the fence required money.

The investigation weighed these arguments and accepted influence on the balance of the evidence. The deciding consideration was that a documented plan brings an audit trail and an identified owner. Every other control that failed in this occurrence failed for want of exactly those two things. The counter-argument was not dismissed. This finding is held with less confidence than the other contributing factors in this report, and it should be read on that basis.

The threshold sat upstream of two other contributing factors, which are analysed under their own headings. The first is the deferral of the perimeter fence repair for three budget cycles. The second is the requirement for a runway inspection only at listed aerodromes, where the listing criterion is a strike.

The size threshold in AS-139 10.4.2 left Tarlton Springs, lawfully, without a documented wildlife hazard management plan, and the absence of a plan likely contributed to the occurrence by leaving the aerodrome's knowledge of kangaroo activity and of the failed fence without an owner or a path to the operator.

Organisational Influences

Perimeter fence repair deferred for three budget cycles

The aerodrome operator's own works register shows that the northern boundary fence had failed, and was known to have failed, more than three years before the occurrence. Item WR-221 was raised in 04/2023. It described four displaced panels and strained wire between the 700 m and 1,300 m runway markers, noted that animals had been seen passing through, and recommended replacement at an estimated \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds. Each deferral recorded the work as unfunded and put it back to the next cycle. Item WR-263, raised in 02/2026, noted that the same section had deteriorated further. None of this had to be reconstructed.

The site inspection on 23/06/2026 showed the condition of the fence on the ground. A continuous 140 m section had its bottom two wires separated from the posts, and a clear animal pad ran through it at the 1,050 m mark (Figure 1). The aircraft was rolling on runway 09, and the first officer saw the animals moving onto the runway from the northern side, past the 1,100 m point. The two carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold. The breach, the pad, the direction from which the animals came and the strike position all lie along the same northern side of the runway, within a short distance of one another.



Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026

The aerodrome's wildlife sighting log shows that kangaroos were present throughout the period of deferral. Between 01/2025 and 06/2026 it held 46 entries. Of these, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 fell between one hour before and two hours after last light. Eleven entries recorded animals on the sealed surface of runway 09/27.

A fence in the condition found does not exclude kangaroos. The animal pad through the breach led directly towards the strike position. The repair was deferred under a regulatory arrangement that required no documented wildlife hazard management plan at an aerodrome of this size. That arrangement is discussed under External Influences.

Deferring the northern perimeter fence repair across three budget cycles very likely left open the route by which the two kangaroos reached the runway.

Evening service approved as a schedule change, route risk assessment not reopened

The current route risk assessment for Tarlton Springs was RRA-041 issue 3, dated 11/2022. Among the hazards it assessed were "runway length and accelerate-stop margin" and "single runway with no lighting redundancy". It carried the operating restriction "daylight operations only pending review of night lighting and diversion fuel policy".

Change Management Record CMR-2025-118 introduced a scheduled evening rotation from 14/07/2025. This was the first company operation into Tarlton Springs after last light. The change was assessed as "schedule change only. Existing aerodrome, existing aircraft type, existing crew

qualifications, existing engineering support arrangements. No new hazard identified." It answered "No" to whether a route risk assessment review was required. The Head of Flying Operations approved it on 22/05/2025.

The daylight-only restriction is not referenced anywhere in CMR-2025-118. RRA-041 was not reopened, and no issue 4 exists. The Safety Department's involvement was limited to receiving the approved change. These are documentary facts and nothing contradicts them. It is virtually certain that the evening service was approved as a schedule change and that the route risk assessment was not reviewed.

Why the change was assessed in this way is not known. The approver has left the company and could not be interviewed. This analysis therefore draws no conclusion about intent. What the documents do show is the question the process asked. It asked whether the aerodrome, aircraft, crews and support were new. It did not ask whether the existing assessment still held for an operation it had expressly excluded.

The influence of that omission rests on what a review would have reached. RRA-041 was a route-level assessment of Tarlton Springs. A review of it against a night operation would very plausibly have reached the aerodrome's wildlife exposure. In the same pass it would have reached the operator's wildlife-active aerodrome listing, which governed whether a runway inspection was required before the first departure after last light.

Two considerations weigh against that. The restriction named only night lighting and diversion fuel, so a review might have addressed those matters alone. And the approver could not be asked what information was in front of him. These considerations keep the influence short of certain. The inspection requirement, and its dependence on a reported strike, is argued under Risk Controls.

Approving the evening service as a schedule change without reopening RRA-041 likely allowed night operations at Tarlton Springs to begin without the aerodrome's wildlife exposure being assessed against the operator's runway inspection requirement.

Risk Controls

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator had a defence aimed at this hazard. At an aerodrome listed in Appendix 8A, Part A 8.14.2 required the pilot in command to "request a runway inspection from the aerodrome operator prior to the first departure after last light". The crew could not then commence the take-off roll until the result had been passed. At every other aerodrome, 8.14.3 stated that "no specific action is required."

The weakness lay in how an aerodrome came to be listed. Appendix 8A admitted an aerodrome only where "a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." No provision in Part A asked a crew to report a sighting. No provision admitted an aerodrome on any other evidence. The control could therefore be armed only by the event it existed to prevent. Tarlton Springs had never had a company strike and had never been listed. The list was last reviewed on 03/03/2025, four months before the evening service began.

The local exposure was documented, but it did not reach the operator. The aerodrome's paper sighting log for 01/2025 to 06/2026 held 46 entries, and 41 of them were kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light. Eleven recorded animals on the sealed surface of runway 09/27. The log was not reported

to any operator, and no airline had asked to see it. Under the listing criterion, none of this could have placed the aerodrome on the list.

It was virtually certain that the requirement existed in this form and did not apply at Tarlton Springs. Had it applied, an inspection would have been requested and the roll would not have begun until a result was passed. Aerodrome staff disperse animals when they find them. However, an inspection completed some minutes before departure would not have guaranteed a clear runway at the roll, and that limits the weight the counterfactual can bear. The absence of an inspection on the night is examined under Individual Actions.

It was very likely that the strike-based listing criterion contributed to the occurrence, because it left Tarlton Springs outside the inspection requirement and so contributed to no inspection being requested before the departure.

Rejected take-off training covers aircraft system failures below V1 only

Part B 4.6 set the frame against which the decision was made. Under 4.6.4, "At or above V1 the take-off shall be continued." The section gave no guidance on an obstruction seen on the runway at any speed. It also gave none for a case in which continuing was itself hazardous.

The training matched the manual. Module 4 of TRG-SYL-07 had four scripted scenarios: engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt, and engine failure at V1 plus 5 kt, continued. All four were aircraft system events. In the five cycles held on file, no crew had been presented with an external obstruction. In the 2025 to 2026 cycle alone, 214 crew completed the module. Both pilots stated independently that every stop or go scenario they had been given was an engine event arriving below V1. It was extremely likely that this gap existed. The qualification reflects only that scenario records before the five cycles on file were not available.

Whether the gap influenced this reject was argued both ways. The first officer's call came at 07:30:31 and the captain's decision at 07:30:32. On one reading, the captain met the question for the first time in that interval, with no briefed or practised frame behind him. On the other reading, the captain stated that he knew the rule and departed from it deliberately because he judged the alternative worse. Obstruction training would not have changed what he knew. On that view the gap is a finding about the fleet rather than a cause of this reject. The investigation weighed both readings. An unrehearsed decision made in about two seconds is not the same decision as one made against a briefed and practised frame, but the captain's account keeps the second reading open. This explains the conditions in which the decision was made. It does not substitute for the decision, which is examined under Individual Actions.

It was likely that the absence of any rejected take-off training or guidance for an obstruction on the runway contributed to the take-off being rejected 6 kt above V1, although the captain's account supports the alternative reading that the gap was a fleet-wide matter rather than a cause of this reject.

Cabin floor proximity emergency lighting partially unserviceable

The first officer noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. A functional check on 24/06/2026 confirmed the defect. With the system armed and on battery power, two of the six floor proximity escape path segments aft of row 16 on the left side were dark. The fault was traced to a failed ballast common to both segments. It was virtually certain that the defect existed.

It was unlikely to have had any effect on the overrun or its outcome. No evacuation was commanded and no slides were deployed. Passengers left by the forward airstair without urgency, with the main cabin lights on throughout.

The defect matters for what it shows about the defence. The pre-flight test on this type illuminated the ceiling emergency lights and exit signs. It neither illuminated nor monitored the floor segments. The failure was therefore undetectable by the crew or a line engineer without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month programme interval. The same ballast is fitted across the type, so the state of the fleet is unknown. In a night evacuation with a fire indication or smoke, two dark segments in the aft cabin could be the difference between an orderly exit and a blocked one.

It was very likely that the undetectable failure of the floor proximity lighting, monitored only at a 24-month interval, increased risk to occupants in any evacuation, although it was unlikely to have had any effect on this occurrence.

Local Conditions

Departure in the hour after last light, the peak of kangaroo movement

The take-off roll began at 17:30 local. That was 32 minutes after sunset and 5 minutes after the end of civil twilight. Moonrise was not until 22:41, and the moon was 14% illuminated. There was no natural ambient light and no town lighting within sight, so nothing lit the graded strip either side of the sealed surface. The investigator's site visit at the same time of day on 23/06/2026 confirmed that the strip cannot be seen from the centreline.

The departure also fell in the period of greatest wildlife movement at the aerodrome. The aerodrome's sighting log holds 41 kangaroo entries. Of those, 38 fall between one hour before and two hours after last light, and eleven record animals on the sealed surface of runway 09/27. The log is a paper register, and its completeness could not be verified. That limits confidence in the exact distribution but not in its general pattern.

The light condition bore directly on detection. Shortly after the stop, the first officer said the animals "came out of the dark on the left of the strip", and that "There was nothing to see and then there were two of them." Neither pilot could see the animals until they were on the lit surface. Darkness on its own is common to every night departure. Here it coincided with the recorded peak of kangaroo movement, and that combination is what gave the absence of a runway inspection its significance. The inspection requirement is argued under Risk Controls.

It was extremely likely that the departure took place in full darkness during the recorded peak of kangaroo movement, and likely that this condition contributed to two kangaroos entering the runway strip and moving onto the sealed surface unseen.

Runway edge lighting intensity reduced the crew's ability to detect the animals

The investigation examined whether the intensity of the runway edge lighting reduced the crew's ability to see the animals. This could not be resolved.

The intensity itself was measured. A photometric check on 25/06/2026, seven days after the occurrence, found four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3. This was the section the animals crossed. The remaining lights were within tolerance. The aerodrome's maintenance programme checks lamp serviceability but does not measure intensity.

The captain's account supported an effect. He recalled the lighting as brighter than he would have wanted, a corridor of light with everything outside it black. He said anything off the seal could not be seen approaching.

The first officer's account did not support an effect, and she was the pilot who saw the animals. She described the lighting as normal. She said she saw the animals clearly once they were on the

sealed surface, at about 200 m. She also said that her two previous visits to the aerodrome had both been in daylight, so she was not well placed to judge whether the lighting was unusual. The captain's not seeing the animals does not need the lighting to explain it. At 07:32:47 he said "I was heads-in on the engine page after V one", which is where the procedure placed him.

The measurement shows a deviation but not an effect. No measurement exists of the lights' output on the night. The two accounts disagree. Nothing establishes a link between an edge light at 131% of nominal and the detection of an object on the runway. Settling the matter would need evidence of the lighting's output on the night and evidence linking edge light intensity of that order to detection. Neither was available.

Whether the lighting reduced the crew's ability to detect the animals was about as likely as not, and the matter remains unresolved. Nothing from it is carried into the findings.

Take-off commenced with a tailwind above limits

The investigation tested whether the take-off began with a tailwind above limits. The observations and the FDR, which are independent of each other, both recorded a headwind on runway 09 throughout the roll, and the crew's performance entry used a 5 kt headwind. It was very unlikely that a tailwind existed, and the hypothesis was excluded.

Fifth sector, 11 hours into duty, 40 minutes behind schedule

The crew signed on at 06:30 local. The occurrence was at 17:30, 11 hours into a five-sector duty planned to finish at 18:35. That gave a duty of 12 hours 05 minutes against a limit of 12 hours 30 minutes. The fifth sector was added that morning, when the rostered crew reported unfit, and the aircraft departed 40 minutes behind schedule. Both pilots described themselves as tired but not struggling. The captain said he would have declined a further sector, but that he was fully alert in the seconds that mattered.

A late fifth sector at the end of a long duty is where degraded decision making would be expected. The recorded data do not show it. The power levers reached flight idle 0.6 seconds after the decision, and maximum braking followed at 2 seconds against a fleet mean of 3.2 seconds. There was no error of omission on the roll. The one decision the duty might have affected was a runway inspection request. No document prompted either pilot to consider one, so fatigue cannot be found to have caused the omission of a step neither pilot was asked to take.

It was virtually certain that the crew were operating a late fifth sector 11 hours into their duty, and unlikely that this contributed to the occurrence. It was likely that the arrangement increased risk regardless. The evening rotation places the last sector near the end of a duty of about 12 hours, at an aerodrome with no company engineering, no alternate within 200 nm at night, and no crewing resilience. A single unfit crew member turns a four-sector day into a five-sector day on the morning of operation.

Individual Actions

No runway inspection requested before the first departure after last light

Neither the CVR nor the tower recorded audio contains a request for a runway inspection, an offer of one, or any mention of wildlife before departure. The aerodrome's daily inspection log for 18/06/2026 shows one inspection, at 07:35 local, before the first arrival. It found the movement area clear, and no later inspection was carried out or requested. It is virtually certain that no inspection was requested before SB512 departed.

Fictional case study for training. Not a real investigation.

The inspection was available to the crew. Aerodrome Manual 6.3 provides for an inspection outside the scheduled time at the request of a pilot in command. The reporting officer confirmed that one would have been carried out if asked.

Neither crew member considered asking for one. Part A 8.14.2 requires the request only at an aerodrome listed in Appendix 8A, and 8.14.3 states: "At all other aerodromes, no specific action is required." Tarlton Springs was not listed. The captain had expected it to be listed and was surprised to find it was not. Why the listing criterion left Tarlton Springs off the list is argued under Risk Controls. No document prompted either crew member to make the request. The omission came from that gap, not from a required step left undone.

An inspection would have covered the section from 1,050 m to 1,250 m, where the fence pad and both carcasses were found. Aerodrome staff disperse animals when they find them. However, an inspection some minutes before the roll could not guarantee that the runway would stay clear.

It is likely that the absence of a runway inspection contributed to the two kangaroos being on the sealed surface during the take-off roll.

Take-off rejected 6 kt above V1

The FDR records V1 at 112 kt at 07:30:29. The first officer called the kangaroos at 07:30:31. At 07:30:32 the captain said "Stopping. I have control." Both power levers reached flight idle within 0.6 seconds, at 118 kt. The reject therefore began three seconds and six knots after V1, and the captain's own account agrees. It is virtually certain that the take-off was rejected above V1.

Part B 4.6.4 states: "At or above V1 the take-off shall be continued." Section 4.6 gives no guidance on an obstruction seen on the runway at any speed. It also says nothing about a case in which continuing is itself hazardous.

The decision was made and announced in a single exchange. Afterwards the captain told the first officer "I never had them." He acted on the first officer's call alone. In interview and in his air safety report, the captain explained his reasoning. He judged that rotating into two large animals would leave the aircraft airborne at night with damage he could not assess, over country with no usable option. He chose the risk whose end he could see, and he acknowledged that the decision was outside the procedure.

Two related matters are dealt with elsewhere. Whether training prepared crews for this choice is argued under Risk Controls. Whether the edge lighting affected how early the animals could be seen remains unresolved under Local Conditions, and nothing from it is carried here.

The scheduled accelerate-stop distance of 1,780 m assumes the stop begins at or before V1. The margin between the distance required and the distance available was 67 m. The reject began 1,152 m along runway 09 with 695 m of seal remaining. The stop actually achieved from 118 kt took 757 m. Beginning the stop three seconds and six knots late used up the 67 m margin and more, before the loss of No. 2 reverse is counted.

The decision followed from the animals being on the runway and from the training provided. It is one contributing factor among several.

It is extremely likely that rejecting the take-off above V1 contributed to the aircraft being unable to stop on the sealed surface.

Maximum braking and No. 1 reverse applied without delay

The stop itself was executed quickly. Maximum braking was established at 07:30:34, two seconds after the decision. The operator's fleet mean for rejected take-offs is 3.2 seconds from decision to

maximum braking. Anti-skid was active on all four wheels. Brake energy reached 84% of the certified maximum without a brake fire.

No. 1 reverse was selected at 07:30:36. The strike had damaged the No. 2 propeller, so No. 2 reverse was not available. The captain identified this immediately with the call "number two's not coming". The first officer called the speed decay and the approaching runway end.

The aircraft left the seal at 21 kt. It stopped 62 m into the 90 m RESA, upright and on the extended centreline, and was disembarked without an evacuation. A stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m, with 28 m of the RESA remaining.

The reject began outside the procedure, but the stop that followed was performed faster than the fleet average.

It is very likely that the prompt application of maximum braking and No. 1 reverse reduced how far the aircraft travelled beyond the runway end.

Occurrence Events

Two kangaroos entered the runway strip and moved onto the sealed surface

It is virtually certain that two kangaroos were on the sealed surface of runway 09 during the take-off roll. Three independent sources each establish this on their own:

- two adult carcasses were recovered from the runway at 1,188 m and 1,221 m from the threshold
- the first officer called the animals on the CVR at 07:30:31
- an impact was recorded at 07:30:33.

The first officer saw the animals crossing from the northern side towards the south. The estimated range at first sight was around 200 m. Both carcasses lay on the northern half of the runway. The animal pad through the failed northern fence is at the 1,050 m mark, 138 m before the first carcass on the same side. The positions are consistent with the animals coming onto the strip through that gap. What this shows about the deferred fence repair is argued in the analysis of organisational influences.

The presence of the animals was not unusual for the aerodrome. Its sighting log for 01/2025 to 06/2026 contains eleven entries of kangaroos on the sealed surface of runway 09/27. Most kangaroo entries in the log fall around last light. The departure took place in the hour after last light, the period of peak kangaroo movement, and that condition is argued in the analysis of local conditions.

The animals' presence led directly to the two events that followed. The captain's words "Stopping. I have control." came one second after the sighting call, and no other cue preceded them. The strike followed because the animals were in the aircraft's path.

It is extremely likely that the kangaroos' presence on the sealed surface led to both the rejected take-off and the strike.

No. 2 propeller and right main gear struck two kangaroos

It is virtually certain that the aircraft struck both animals. The FDR recorded a lateral acceleration spike of 0.31 g right and a vertical spike of 1.48 g at 07:30:33, at 114 kt and one second after the reject began. Three No. 2 propeller blades were damaged beyond repairable limits, and one blade tip had separated. Animal tissue was recovered from two blade roots. The right main gear doors were distorted, and propeller debris punctured the lower right fuselage skin.

The strike's effect on the stop was immediate. No. 2 propeller RPM decayed from the moment of impact, and the No. 2 reverse interlock did not make. When the captain selected reverse at 07:30:36, he called that "number two's not coming", and only No. 1 reverse was established. The aircraft therefore decelerated for the rest of the stop with half its reverse thrust.

The investigation found no other cause for the reduced deceleration. Maximum braking was held at 2,940 psi with anti-skid active on all four wheels. The anti-skid system tested serviceable after the occurrence, and the brakes showed no fire or hydraulic leak. The right main tyres deflated through the fuse plugs, which is consistent with brake energy rather than the impact. The gear structure was within limits.

It is virtually certain that the strike removed No. 2 reverse early in the stop, which reduced the deceleration available when the runway ahead had no margin to absorb the loss.

Deceleration achieved below the scheduled accelerate-stop performance

The scheduled accelerate-stop distance required was 1,780 m, against an ASDA of 1,847 m. That left a margin of 67 m. The figure rested on two assumptions: that the stop began at or before V1, and that both reverse units were available. Neither assumption held on this occasion.

Distance reconstruction from recorded groundspeed placed the aircraft 1,152 m along runway 09 when the reject began at 118 kt. That left 695 m of sealed surface. The stopping distance actually achieved was 757 m. Mean deceleration under maximum braking was 2.6 m/s², approximately 15% below the value used in the certified data for this weight and condition.

The aircraft left the sealed surface at 21 kt. In the gravel, deceleration fell to 1.9 m/s². The aircraft stopped 62 m beyond the runway end, on the extended centreline, and the nose wheel rut in the RESA matches that position.

The 62 m overrun is the product of two contributions acting together. The first was a stop begun 6 kt above V1, which is argued in the analysis of individual actions. The second was deceleration about 15% below the scheduled value, which followed from the strike. The investigation found that both were needed to carry the aircraft beyond the runway end, and that neither alone would have done so.

It is virtually certain that a late start to the stop, combined with deceleration below the scheduled performance, carried the aircraft 62 m beyond the end of runway 09.

No. 2 engine power loss during the take-off roll

A No. 2 engine power loss during the take-off roll was very unlikely. Both engines held take-off torque within 0.1% of each other, with no caution, warning or exceedance, until the power levers were retarded at 07:30:32, and the No. 2 RPM decay began only at the impact one second later.

Conclusions

These findings are made with respect to the occurrence and should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. AS-139 10.4.2 required a documented wildlife hazard management plan only at aerodromes used by aeroplanes above 27,000 kg MTOW or serving more than 350,000 passenger movements a year. Tarlton Springs met neither threshold. It held no documented plan, hazard assessment or dispersal programme, but it was not in breach. It is virtually certain that this condition existed. It is likely that it contributed, because nothing connected the aerodrome's sighting log to operators or placed the failed fence on a funded programme. That influence, however, was more finely balanced than for the other organisational findings. A documented plan was not the only means of repairing a fence or sharing a sighting log, and there was no evidence that a plan would have been funded any better. (Safety issue)

2. Works item WR-221 was raised in 04/2023 and recorded animals passing through the failed northern perimeter fence. At the 2023, 2024 and 2025 budget rounds, the repair was deferred unfunded. By 23/06/2026 the fence had a continuous 140 m failed section. An animal pad ran through it at the 1,050 m mark, on the same side of the runway as the carcasses at 1,188 m and 1,221 m. It is virtually certain that the repair was deferred for three budget cycles. It is very likely that the deferral contributed, by leaving a route for kangaroos onto the runway strip. (Safety issue)

3. CMR-2025-118 introduced a scheduled evening service into Tarlton Springs from 14/07/2025 and assessed it as a schedule change only. As a result, it did not reopen RRA-041 issue 3, which restricted the aerodrome to daylight operations pending review of night lighting and diversion fuel policy. It is virtually certain that the change was approved in this way. It is likely that this contributed, because a review against a night operation could have reached the aerodrome's wildlife exposure and the Appendix 8A listing. There were two reasons the influence was no higher than likely. First, a review might have addressed lighting and fuel alone, since those were what the restriction named. Second, the approver could not be interviewed. (Safety issue)

4. Operations Manual Part A 8.14.2 required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A. The only criterion for listing was a wildlife strike involving a company aircraft reported at that aerodrome in the preceding 24 months. Tarlton Springs had never been listed, even though its sighting log recorded eleven occasions on which kangaroos were on the sealed surface. It is virtually certain that the inspection requirement was conditional in this way. It is very likely that this contributed, because it left the operator's inspection requirement inapplicable on the evening of the occurrence. Even so, an inspection completed some minutes before the roll could not have guaranteed a clear runway. (Safety issue)

5. All four rejected take-off scenarios in TRG-SYL-07 module 4 were aircraft system events, and three of them were below V1. No crew had been given a runway obstruction scenario in the five training cycles on file. Nor did Operations Manual Part B 4.6 contain any guidance on an obstruction on the runway at any speed. It is very likely that this gap existed. It is likely that it contributed, because the pilot in command made an unrehearsed decision in the second between the sighting call and the rejection. The pilot in command, however, stated that he knew the rule and departed from it deliberately. On that reading, the gap is a deficiency across the fleet rather than a cause of this rejection. The evidence supports both readings. (Safety issue)

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6. On 24/06/2026 a functional check found two of the six floor proximity escape path segments aft of row 16 on the left side dark on battery power with the system armed. The fault was traced to a failed common ballast. On the type, the pre-flight emergency lighting test neither illuminated nor monitored these segments. The 24-month functional test interval, for its part, could not detect such a defect between tests. It is virtually certain that the defect existed. It did not contribute to the occurrence, because no evacuation was commanded and the main cabin lights were on throughout. Nevertheless, it is very likely that it increased risk, because the same ballast was fitted across the type. (Safety issue)

7. The take-off roll began at 17:30 local. This was 32 minutes after sunset and 5 minutes after the end of civil twilight, before moonrise. At that time the unlit graded strip either side of the sealed surface could not be seen from the centreline. The aerodrome's sighting log placed 38 of its 41 kangaroo entries between one hour before and two hours after last light. It is very likely that the departure fell within this period of peak kangaroo movement. It is likely that this contributed, because the animals could not be seen until they were on the lit surface.

8. The aerodrome observations gave a headwind component of 5 to 7 kt on runway 09, and the FDR recorded a headwind component of 4 to 6 kt throughout the roll. It is very unlikely that the take-off commenced with a tailwind above limits.

9. The occurrence sector was the fifth of the duty, and it had been added that morning when the rostered crew reported unfit. It was flown 11 hours 00 minutes into a duty planned at 12 hours 05 minutes, against a limit of 12 hours 30 minutes, and 40 minutes behind schedule. It is virtually certain that these conditions existed. They did not contribute to the occurrence. The time from decision to maximum braking was at the fast end of the fleet distribution, and there was no error of omission during the roll. Moreover, no document had prompted either pilot to request a runway inspection. It is likely, however, that this duty pattern increased risk. The evening rotation routinely placed the last sector in the last hour of a 12-hour duty, at an aerodrome with no alternate within 200 nm at night.

10. It is virtually certain that no runway inspection was requested before the first departure after last light. The aerodrome would have carried one out if asked, but no document prompted either pilot to consider requesting one. It is likely that the absence of an inspection contributed. An inspection would have covered the section between 1,050 m and 1,250 m, where the fence pad and both carcasses were found, and aerodrome staff dispersed animals when they found them. Even so, an inspection some minutes before the roll could not have guaranteed that the runway stayed clear.

11. The take-off was rejected at 07:30:32 at 118 kt, which was 6 kt and 3 seconds after V1 and one second after the first officer called the animals. Part B 4.6.4 stated that the take-off shall be continued at or above V1. The pilot in command judged that rotating into two large animals would leave the aircraft airborne at night, with damage he could not assess and no usable option. It is virtually certain that the take-off was rejected above V1. It is very likely that this contributed. The scheduled accelerate-stop distance of 1,780 m against an ASDA of 1,847 m left a margin of 67 m, and that margin applied only to a stop begun at or before V1.

12. The power levers reached flight idle 0.6 seconds after the decision. Maximum braking was established 2 seconds after it, against a fleet mean of 3.2 seconds, and No. 1 reverse was selected within 4 seconds. It is virtually certain that maximum braking and No. 1 reverse were applied without delay. It is very likely that this reduced risk. Had the stop begun at the fleet mean response, it would have carried the aircraft an estimated further 30 to 40 m, close to the far edge of the RESA.

13. Two adult kangaroos entered the runway strip from the northern side and moved onto the sealed surface of runway 09. They were recovered at 1,188 m and 1,221 m from the threshold. It is

virtually certain that this occurred. It is very likely that it contributed, because both the rejection and the strike followed directly from the animals being on the surface.

14. At 07:30:33, at 114 kt and one second after the rejection began, the No. 2 propeller and the right main gear struck both kangaroos. Three No. 2 propeller blades were damaged beyond limits, and the No. 2 reverse interlock did not make. It is virtually certain that the strike occurred. It is also virtually certain that it contributed, by removing reverse thrust on one side during a stop with a scheduled margin of 67 m.

15. The rejection began 1,152 m along the runway, with 695 m of sealed surface remaining. From 118 kt, the stopping distance achieved was 757 m. With reverse on No. 1 only, the mean deceleration of 2.6 m/s² was approximately 15% below the certified accelerate-stop value. It is virtually certain that the deceleration fell below the scheduled accelerate-stop performance. It is virtually certain that this resulted in the aircraft stopping 62 m into the RESA. Neither the late rejection nor the loss of No. 2 reverse alone would have carried the aircraft beyond the runway end.

16. Until the power levers were retarded at 07:30:32, both engines held take-off torque within 0.1% of each other, with no caution, warning or exceedance. The No. 2 propeller RPM decay began at impact at 07:30:33. It is very unlikely that a No. 2 engine power loss occurred during the take-off roll.

Probable cause

The aircraft overran runway 09 because the stopping distance it achieved exceeded the runway remaining. This very likely resulted from two factors acting together. The take-off was rejected 6 kt above V1 when two kangaroos crossed the sealed surface after last light, and the No. 2 propeller then struck the animals and removed reverse thrust on that side.

Contributing factors

- It is virtually certain that a documented wildlife plan was required only above a size threshold the aerodrome does not meet. It is likely that this contributed, although that influence was more finely balanced than for the other organisational factors. (Safety issue)
- It is virtually certain that the perimeter fence repair was deferred for three budget cycles, and very likely that the deferral contributed. (Safety issue)
- Departure in the hour after last light, the peak of kangaroo movement, very likely existed and likely contributed.
- Approval of the evening service as a schedule change, without the route risk assessment being reopened, is established with virtual certainty. It is likely that this contributed. (Safety issue)
- A runway inspection was required only at listed aerodromes, with a strike as the listing criterion. This is virtually certain, and the condition very likely contributed. (Safety issue)
- It is virtually certain that no runway inspection was requested before the first departure after last light, and likely that this contributed.
- Two kangaroos entered the runway strip and moved onto the sealed surface. Their presence is virtually certain, and it very likely contributed.
- Rejected take-off training covering aircraft system failures below V1 only very likely existed. It likely contributed, although the pilot in command's account supports a reading in which it did not. (Safety issue)

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- It is virtually certain that the take-off was rejected 6 kt above V1, and very likely that this contributed.
- The No. 2 propeller and right main gear struck two kangaroos. Both the occurrence of the strike and its contribution are virtually certain.
- It is virtually certain that deceleration was achieved below the scheduled accelerate-stop performance, and virtually certain that this contributed.

Other factors that increased risk

- It is virtually certain that the cabin floor proximity emergency lighting was partially unserviceable. This did not contribute to the occurrence, but it very likely increased risk. (Safety issue)
- It is virtually certain that the flight was the fifth sector, 11 hours into duty and 40 minutes behind schedule. This did not contribute to the occurrence, but it likely increased risk.

Other findings

- The hypothesis that the take-off commenced with a tailwind above limits was examined and ruled out, because it is very unlikely that any tailwind component existed during the roll.
- A No. 2 engine power loss during the take-off roll was examined and ruled out as very unlikely, because the No. 2 loss began at impact, after the rejection.
- It is virtually certain that maximum braking and No. 1 reverse were applied without delay, and very likely that this reduced risk.

The investigation also examined whether runway edge lighting intensity reduced the crew's ability to detect the animals. The matter could not be resolved, and no finding is made on it. The evidence for and against an effect was about as likely as not.

On the side of an effect, a photometric check on 25/06/2026 measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3. That section is where the animals crossed. The aerodrome's maintenance programme checked lamp serviceability but never measured intensity. The pilot in command recalled the lighting as brighter than he wanted. He described a corridor of light with darkness either side, and said that nothing off the seal could be seen coming.

Against an effect, the first officer saw the animals and described the lighting as normal. She said that she saw them clearly once they were on the sealed surface, at about 200 m. She added, however, that with only daylight experience at the aerodrome she was not well placed to judge. As for the pilot in command, he was looking at the engine page after V1, as the procedure required. His not seeing the animals therefore did not depend on the lighting.

The check showed an intensity deviation, but it did not show an effect on detection. It was taken seven days after the occurrence, and no measurement exists of the lighting on the night. Nor did the investigation find any evidence linking an edge light at 131% of nominal to the detection of an object on the runway. Settling the matter would have required a measurement of the lighting at the time of the occurrence, and none exists.

Recommendations

Documented wildlife plan required only above a size threshold the aerodrome does not meet

The size threshold in AS-139 10.4.2 meant that Tarlton Springs had to take reasonable measures against wildlife but did not have to document them in a wildlife hazard management plan. Its heaviest scheduled aeroplane weighed 22,900 kg, and it handled 61,400 passenger movements in 2025. Both figures fell below the threshold. The aerodrome was not in breach of the standard. The issue lies in the threshold itself, which left an aerodrome of this size with a wildlife programme that was undocumented, unaudited and, in practice, unfunded.

The issue matters because a documented plan brings an owner and an audit trail. At Tarlton Springs, the sighting log held 41 kangaroo entries in a paper register that no operator had ever asked to see. The failed northern fence remained on an unfunded works register for three budget cycles. The absence of a plan likely contributed to the occurrence. That finding is held with less confidence than the other contributing factors. A plan is not the only means of repairing a fence or passing a sighting log to an operator. There was also no evidence that a documented plan would have secured funding for the repair.

No safety action has been recorded against this issue. No recommendation is made.

Perimeter fence repair deferred for three budget cycles

The aerodrome operator deferred repair of the northern perimeter fence as unfunded at the 2023, 2024 and 2025 budget rounds. The failure was first entered in the works register in 04/2023 as four displaced panels and strained wire between the 700 m and 1,300 m runway markers, with animals observed passing through. The register recorded further deterioration in 02/2026. By the site inspection on 23/06/2026 the failure had become a continuous 140 m section. The bottom two wires had separated from the posts, and an animal pad ran through the fence at the 1,050 m mark (Figure 1).

A fence in that condition does not exclude kangaroos. The breach was very likely the route by which the two kangaroos reached the runway. It lay on the same side of the runway as the carcasses recovered at 1,188 m and 1,221 m. The hazard had been documented in the aerodrome operator's own register, in its own words, for three years before the occurrence.

The aerodrome operator funded works register item WR-221 out of the 2026 contingency. It programmed replacement of the 140 m northern boundary section for 09/2026 and installed a temporary electric wire on 25/06/2026. This action was in progress. [NOT ESTABLISHED: whether the 09/2026 fence replacement was completed.]

The action addresses the breach on the northern boundary. It does not address how a known safety-related defect came to be deferred unfunded across successive budget rounds. No safety action has been recorded on that part of the issue.

Evening service approved as a schedule change, route risk assessment not reopened

The operator approved a scheduled evening rotation into Tarlton Springs as a schedule change, and the route risk assessment for the aerodrome was not reviewed. RRA-041 issue 3, the current route risk assessment, restricted operations at Tarlton Springs to daylight "pending review of night lighting and diversion fuel policy". CMR-2025-118 introduced the first company operation into the

aerodrome after last light. It did not reference the restriction, and RRA-041 was not reopened. It is virtually certain that the change was approved in this way.

The change process asked whether the change was new. It did not ask whether the existing assessment still held. As a result, a company aircraft was scheduled into an aerodrome after last light for the first time, against a live restriction in the operator's own current documentation.

It is likely that a review of RRA-041 against a night operation would have reached the aerodrome's wildlife exposure and the Appendix 8A listing in the same pass. This is not certain. A review might have addressed only lighting and fuel, which are the matters the restriction names. The approver has left the company and could not be interviewed, so the reasons the change was assessed this way are not known.

No safety action has been recorded against this issue, and no recommendation is made.

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator's Operations Manual Part A required the pilot in command to request a runway inspection before the first departure after last light, and not to begin the take-off roll until the result was passed. That requirement applied only at aerodromes listed in Appendix 8A. An aerodrome could be listed only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision asked crews to report a sighting, and no other evidence could admit an aerodrome to the list. Tarlton Springs had never been listed.

The requirement was the operator's own defence against this kind of occurrence, and at Tarlton Springs it did not apply. Its absence very likely contributed to the departure proceeding without a runway inspection. The aerodrome's sighting log held eleven entries of kangaroos on the sealed surface. None of them was a strike, and none was reported to the company. Because the criterion can be met only by the occurrence the control exists to prevent, no aerodrome can be listed before a strike has happened there.

On 26/06/2026 the operator amended Appendix 8A by temporary revision TR-2026-04 to list Tarlton Springs. The inspection requirement therefore applies to every departure there after last light, pending a permanent change. This action addresses the issue in part. It covers Tarlton Springs only and leaves the listing criterion unchanged for every other aerodrome.

The investigation recommends that the operator replace the Appendix 8A listing criterion so that an aerodrome is listed on wildlife exposure rather than on a reported strike. It also recommends that the operator add a route risk assessment trigger for any aerodrome first operated into after last light.

Rejected take-off training covers aircraft system failures below V1 only

The operator's recurrent training prepared crews to reject a take-off only for aircraft system failures below V1. It is extremely likely that this was so. Module 4 of TRG-SYL-07 scripts four scenarios: engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt, and engine failure at V1 plus 5 kt with the take-off continued. In the five training cycles on file, no crew was presented with an external obstruction on the runway. The current cycle alone covers 214 crew. Operations Manual Part B section 4.6 has the same gap. It requires the take-off to be continued at or above V1, and it gives no guidance on an obstruction on the runway at any speed.

The absence of this training likely contributed to the decision to reject the take-off 6 kt above V1. The captain faced the question for the first time and had two seconds to decide. He also stated

that he knew the rule and departed from it deliberately. The gap therefore explains the conditions under which the decision was made. It does not excuse the decision.

The issue reaches beyond this occurrence. Any of the operator's crews meeting an obstruction near or above V1 would face the same decision without a briefed or practised frame.

No safety action has been recorded against this issue, and no recommendation has been recorded.

Cabin floor proximity emergency lighting partially unserviceable

The cabin floor proximity emergency lighting on VH-SBQ was partially unserviceable. The first officer noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. A functional check on 24/06/2026 found two of the six floor proximity escape path segments dark on battery power with the system armed. Both were aft of row 16 on the left side. The defect was traced to a failed common ballast.

It was unlikely that the defect influenced the occurrence. No evacuation was commanded and the main cabin lights were on throughout. The disembarkation, by the forward airstair, took 4 minutes 8 seconds without incident.

The defect was very likely important to safety all the same. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs. It neither illuminates nor monitors the floor segments. As a result, neither the crew nor a line engineer could have detected the defect without a dedicated functional test. The last such test was 11 months before the occurrence, inside the 24-month programme interval.

A fire indication, smoke, or a night evacuation over the wing would have changed the circumstances. In any of those, two dark segments in the aft cabin could be the difference between an orderly exit and a blocked one. The same ballast is fitted across the type. Because the daily test cannot see the floor segments and the functional test interval is 24 months, the true state of the fleet's floor proximity lighting is not known.

No safety action has been recorded against this safety issue.

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